Example:

int x1 = -1, x2 = 1, x3, n;

9.13.3 Dynamic Initialization

A variable can be initialized during the execution of a program. It is known as "**Dynamic initialization**". For example,

int num1, num2, sum;

sum = num1 + num2;

The above two statements can be combined into a single one as follows:

int sum = num1+num2;

This initializes sum using the known values of num1 and num2 during the execution.

Illustration 9.7 C++ Program to illustrate dynamic initialization

```
#include <iostream>
using namespace std;
int main()
{
        int num1, num2;
        cout << "\n Enter number 1: ";
        cin >> num1;
        cout << "\n Enter number 2: ";
        cin >> num2;
        int sum = num1 + num2; // Dynamic initialization
        cout << "\n Average: " << sum /2;
}
```

Output:

Enter number 1: 78
Enter number 2: 65
Average: 71

In the above program, after getting the values of num1 and num2, sum is declared and initialized with the addition of those two variables. After that, it is divided by 2.

Illustration 9.8: C++ program to find the perimeter and area of a semi circle

```
#include <iostream>
using namespace std;
int main()
{
        int radius;
        float pi = 3.14;
        cout << "\n Enter Radius (in cm): ";
        cin >> radius;
        float perimeter = (pi+2)*radius;   // dynamic initialization
        float area = (pi*radius*radius)/2; // dynamic initialization
        cout << "\n Perimeter of the semicircle is " << perimeter << " cm";
        cout << "\n Area of the semicircle is " << area << " sq.cm";
}
```

Output:

Enter Radius (in cm): 14
Perimeter of the semicircle is 71.96 cm
Area of the semicircle is 307.72 sq.cm

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